

The all-clock supremum for universally safe one-site Möbius capacity factors

RH research program

August 2026

Abstract

For every fixed finite clock q , we optimize all phase/current-input sign factors that satisfy the RH-366 distance-two constraint for every possible ternary input word. Their limiting Möbius correlation exists, and its maximum absolute value is the squarefree-density weighted independent-set value

$$F(q) = \max_{I \cap (I+2) = \emptyset} \sum_{r \in I} \delta_{q,r}.$$

The clock need not be a minimal period. Consequently $q \mid Q$ implies $F(q) \leq F(Q)$. We then prove a special saturation law for the RH-374 square clocks $q_y = 4 \prod_{i \leq y} p_i^2$: if $q_y \mid Q$ and Q has the same prime support, then $F(Q) = F(q_y) = B_y$. The proof uses zero phases forced by 4 and 9 to split the two parity cycles; it is not a general cyclic-cover MWIS law. Lifting an arbitrary finite clock to such a same-support multiple gives

$$\sup_{q < \infty} F(q) = B_\infty, \quad F(q) < B_\infty \quad (q < \infty).$$

Thus the RH-374 Euler-product floor is exactly the nonattained all-clock supremum in this one-site class. Memory-dependent observables, growing clocks, infinite selectors, adaptive-capacity convergence, intrinsic operators, traces, Riemann-zero identification, and RH remain outside scope.

1 Frozen capacity and factor class

The RH-366 distance-two capacity is

$$K_N = \max_{\substack{\varepsilon_1, \dots, \varepsilon_N \in \{-1, +1\} \\ \text{not } (\varepsilon_n, \varepsilon_{n+2}) = (+1, +1)}} \left| \sum_{n \leq N} \mu(n) \varepsilon_n \right|. \quad (1)$$

Its optimizer may read the entire finite prefix and may change with N . RH-366 proves the bracket

$$\frac{4}{\pi^2} \leq \liminf_{N \rightarrow \infty} \frac{K_N}{N} \leq \limsup_{N \rightarrow \infty} \frac{K_N}{N} \leq \frac{6}{\pi^2}, \quad (2)$$

but not convergence [6]. RH-371 gives an exact eight-run formula for every prefix while leaving the required Möbius run envelope open [5]. RH-373 and RH-374 instead construct prescribed one-site certificates, culminating in a strictly increasing square-clock family $B_y \uparrow B_\infty$ [4, 7].

Definition 1.1 (One-site factor and universal safety). Fix $q \geq 1$. A q -periodic one-site phase/current-input factor is a family

$$g_r : \{-1, 0, +1\} \longrightarrow \{-1, +1\}, \quad r \in \mathbb{Z}/q\mathbb{Z}.$$

The declared period need not be minimal. For a ternary word (a_n) , its output is $g_{n \bmod q}(a_n)$. The factor is *universally safe* if this output never has two $+1$ values at distance two, for every ternary input word. Its active set is

$$S_g = \{r : g_r(a) = +1 \text{ for at least one } a \in \{-1, 0, +1\}\}.$$

Universal safety is equivalent to

$$S_g \cap (S_g + 2) = \emptyset. \quad (3)$$

Indeed, two active phases at distance two can be activated independently; the converse is immediate. This retains loops: for $q = 1$ and $q = 2$, addition by two fixes every phase, so only the empty active set is safe. The class is the one-site boundary isolated in RH-372, not the class of all finite-memory transducers [3].

2 Exact optimization at one clock

For $r \pmod q$, let

$$\delta_{q,r} = \lim_{N \rightarrow \infty} \frac{1}{N} \#\{n \leq N : n \equiv r \pmod q, \mu(n)^2 = 1\}. \quad (4)$$

The squarefree progression sieve gives this limit. If $q = \prod_{p|q} p^{a_p}$, its useful exact local form is

$$\pi^2 \delta_{q,r} = \begin{cases} 0, & p^2 \mid r \text{ for some } p \text{ with } a_p \geq 2, \\ \frac{6}{q} \prod_{p|q} (1 - p^{-2})^{-1} \prod_{\substack{p||q \\ p \nmid r}} (1 - p^{-1}), & \text{otherwise.} \end{cases} \quad (5)$$

The second product accounts for a modulus containing only one power of p . It is an exact rational coefficient of π^{-2} [2].

For fixed q , Davenport cancellation and finite Fourier inversion give

$$\sum_{\substack{n \leq N \\ n \equiv r \pmod q}} \mu(n) = o(N).$$

Splitting the squarefree entries by sign therefore proves, for every one-site factor,

$$L_q(g) := \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n \leq N} \mu(n) g_{n \bmod q}(\mu(n)) = \frac{1}{2} \sum_{r \bmod q} \delta_{q,r} (g_r(+1) - g_r(-1)). \quad (6)$$

This is the fixed-clock formula proved in the RH-372 certificate layer [1, 3].

Definition 2.1 (Fixed-clock optimum). Let

$$F(q) = \max_g |L_q(g)|,$$

where the maximum is over all universally safe q -periodic one-site factors.

Theorem 2.2 (Weighted phase-MWIS formula). *For every finite q ,*

$$F(q) = \max_{\substack{I \subset \mathbb{Z}/q\mathbb{Z} \\ I \cap (I+2) = \emptyset}} \sum_{r \in I} \delta_{q,r}. \quad (7)$$

Both correlation orientations attain the value. In particular, $F(1) = F(2) = 0$.

Proof. Put $c_r = (g_r(+1) - g_r(-1))/2 \in \{-1, 0, +1\}$. Whenever $c_r \neq 0$, phase r is active, so the support of (c_r) is contained in the independent set S_g . Equation (6) gives

$$|L_q(g)| \leq \sum_{r: c_r \neq 0} \delta_{q,r} \leq \max_{I \cap (I+2) = \emptyset} \sum_{r \in I} \delta_{q,r}.$$

Conversely, for an independent set I , set

$$g_r(+1) = +1, \quad g_r(0) = g_r(-1) = -1 \quad (r \in I), \quad g_r \equiv -1 \quad (r \notin I).$$

This universally safe factor has $L_q(g) = \sum_{r \in I} \delta_{q,r}$. Reversing the roles of $+1$ and -1 on I realizes the negative orientation. The self-loops at $q = 1, 2$ force $I = \emptyset$. $\square$

For odd $q > 1$, addition by two is one q -cycle; for even $q > 2$, it is two cycles of length $q/2$. Thus (7) includes odd clocks rather than silently reducing to the two-parity case used by the square family.

Remark 2.3 (Deletable phases). The value $g_r(0)$ contributes nothing to (6); changing an unnecessary $+1$ there to -1 can only remove a safety constraint. Likewise, an active zero-density phase can be made inactive without changing the objective. These observations do not delete positive-weight phases and do not enlarge the theorem beyond one-site factors.

3 Divisibility monotonicity

Lemma 3.1 (Density aggregation). *If $q \mid Q$, then for every $r \pmod{q}$,*

$$\delta_{q,r} = \sum_{\substack{s \pmod{Q} \\ s \equiv r \pmod{q}}} \delta_{Q,s}. \quad (8)$$

Proof. For every finite N , the squarefree integers in the class $r \pmod{q}$ are the disjoint union of the classes $s \pmod{Q}$ above it. Divide by N and take the finitely many limits. $\square$

Proposition 3.2 (Clock divisibility). *If $q \mid Q$, then*

$$F(q) \leq F(Q). \quad (9)$$

Proof. View a q -periodic factor as the Q -periodic factor $\tilde{g}_s = g_{s \bmod q}$. This is why minimal period was not required. Condition (3) shows that universal safety is preserved. The two factors produce the same output word, hence the same limit; equivalently, this follows by inserting (8) into (6). Maximizing gives (9). $\square$

4 Special square-clock saturation

Let $3 = p_1 < p_2 < \dots$ be the odd primes and recall the RH-374 quantities

$$P_y = \prod_{i \leq y} p_i^2, \quad q_y = 4P_y, \quad A_y = \prod_{i \leq y} (p_i^2 - 1). \quad (10)$$

On the odd phase cycle, let O_y be the number of odd-length positive runs in one period of the word

$$w_y(k) = \mathbf{1}_{\{p_i^2 \mid 2k+1 \text{ for all } i \leq y\}}.$$

RH-374 proves

$$F(q_y) = B_y = \frac{4 + 2O_y/A_y}{\pi^2}, \quad (11)$$

and proves B_y strictly increasing. With

$$e_m = \prod_{p \text{ odd}} (1 - m/p^2), \quad e_9 = 0,$$

its limit is

$$B_\infty = \frac{4 + 2C}{\pi^2}, \quad C = \frac{1}{e_1} \sum_{j \in \{1,3,5,7\}} (e_j - 2e_{j+1} + e_{j+2}). \quad (12)$$

Proposition 4.1 (Same-prime-support saturation). *Suppose $q_y \mid Q$ and Q has exactly the same prime divisors as q_y . Write $Q = Rq_y$. Then*

$$\boxed{F(Q) = F(q_y) = B_y.} \quad (13)$$

Proof. Because every prime in the support occurs to exponent at least two, a phase $s \pmod{Q}$ has positive squarefree density precisely when no supported prime square divides s . All positive phases have the same weight. By (5), it is $1/R$ times the positive q_y weight, hence

$$\delta_{Q,s} = \frac{2}{RA_y\pi^2}. \quad (14)$$

It remains to count the largest independent subset of the positive support. The distance-two graph has one even and one odd phase cycle. On the even cycle, every phase divisible by 4 has zero weight; these zero phases split the support, and all RA_y positive even phases may be selected. On the odd cycle, the support is R repetitions of the q_y support. The factor $p_1^2 = 9$ inserts a zero every nine successive odd-cycle sites, so all positive runs have length at most eight and the repetitions do not join across a hidden all-positive seam. The RH-374 run count therefore repeats literally: its odd-cycle MWIS cardinality is $R(A_y + O_y)$. The total support MWIS cardinality is

$$R(2A_y + O_y). \quad (15)$$

Multiplying (15) by (14) gives $(4 + 2O_y/A_y)/\pi^2 = B_y$. $\square$

Remark 4.2 (No general cover law). An MWIS need not scale under an arbitrary cyclic cover. Proposition 4.1 uses the particular squarefree support: forced 4-zero phases split the even component and forced 9-zero phases split the odd component. No claim is made for a general weighted cyclic graph.

5 The all-clock supremum

Theorem 5.1 (Exact finite-clock one-site supremum). *Over all finite clocks and universally safe one-site phase/current-input factors,*

$$\boxed{\sup_{q < \infty} F(q) = B_\infty.} \quad (16)$$

The supremum is not attained: every fixed finite q satisfies

$$F(q) < B_\infty. \quad (17)$$

Proof. Fix an arbitrary finite q . Choose y so that every odd prime divisor of q occurs among $p_1, \dots, p_y$; when q has no odd prime divisor, take $y = 1$. Put

$$Q = \text{lcm}(q, q_y).$$

Then $q \mid Q$, $q_y \mid Q$, and Q has exactly the prime support of q_y : extra exponents in q , including high powers of 2 or of an odd prime, only enlarge the multiplier $R = Q/q_y$. Propositions 3.2 and 4.1 give

$$F(q) \leq F(Q) = B_y < B_\infty,$$

where the last inequality uses the strict RH-374 sequence. This proves the upper bound in (16) and the nonattainment statement. Conversely $F(q_y) = B_y$ for every y , so $\sup_q F(q) \geq \sup_y B_y = B_\infty$. $\square$

Corollary 5.2 (Relation to adaptive capacity).

$$B_\infty \leq \liminf_{N \rightarrow \infty} \frac{K_N}{N}. \quad (18)$$

Proof. Every universally safe factor supplies an admissible word in (1), so $F(q) \leq \liminf K_N/N$ for each fixed q . Take the scalar supremum and apply (16). $\square$

This recovers the RH-374 floor while adding an exact interpretation: it is the all-finite-clock supremum in the one-site class. It does not identify B_∞ with an ordinary limit of K_N/N and does not choose $q = q(N)$.

6 Exact executable audit

The artifact evaluates (5) with exact `Fraction` arithmetic and solves each addition-by-two cycle by a weighted path/cycle DP. It retains self-loops and handles odd clocks, high 2-adic exponents, high odd-prime exponents, zero weights, and both factor orientations. Its independent finite checks are:

- all 8^q factor tables for $1 \leq q \leq 4$: 4680 total tables and 249 universally safe tables;
- all phase subsets for $1 \leq q \leq 10$: 2046 subsets, with every exact optimum agreeing with the cycle DP;
- eight divisibility pairs, checking every density aggregation row, the lifted independent set, and $F(q) \leq F(Q)$;
- ten cofinal lifts, including $q = 16, 27, 125, 343$ and the largest audited lift $343 \mid 308700$;
- the RH-374 square rows

$$\pi^2 F(36) = 4, \quad \pi^2 F(900) = \frac{49}{12}, \quad \pi^2 F(44100) = \frac{593}{144}.$$

Representative lift rows are

q	Q	R	$\pi^2 F(q)$	$\pi^2 F(Q)$
3	36	1	9/4	4
8	72	2	4	4
27	108	3	3	4
125	4500	5	3	49/12
343	308700	7	3	593/144

A bounded reproduction scan over all $1 \leq q \leq 256$ has record clocks 1, 3, 4, 180 and maximum $\pi^2 F(q) = 97/24$ at $q = 180$. This finite scan is not evidence for (16); the theorem is the cofinal argument in Section 5. The result ledger locks twenty-one repository inputs. The ARS integrity audit is repository-locked: it checks claim/source alignment, artifact provenance, and all seven research failure modes, but makes no fresh external bibliographic-verification claim.

7 Route verdict and Gate ledger

Route A is `G0`: equations (7), (9), (13), and (16) form a standalone exact theorem chain. Route B is `STOP_SCOPED`. The class excludes memory-dependent observations; no higher-order Möbius correlation theorem is supplied. There is no $q(N)$, infinite selector, or Davenport estimate uniform in the clock, and the adaptive capacity limit remains open.

Gates A–E remain false/open. These factors are externally prescribed arithmetic certificates, not intrinsic dynamical operators. This paper does not construct a Fredholm or spectral determinant, a time-oriented unitary completion, a self-adjoint generator, a von-Mangoldt prime-power trace, or a completed-zeta divisor equality. It does not identify Riemann zeros, construct a Hilbert–Polya operator, or prove the Riemann Hypothesis.

8 Conclusion

The finite-clock one-site question has an exact endpoint: the RH-374 Euler-product constant is its supremum, approached by square clocks and attained by no finite clock. The remaining RH-366 problem is genuinely different. It requires control of adaptive or memory-dependent arithmetic structure; another bounded scan or an unproved growing-clock exchange cannot supply that control.

References

- [1] Harold Davenport. On some infinite series involving arithmetical functions (II). *The Quarterly Journal of Mathematics*, os-8(1):313–320, 1937.
- [2] Leon Mirsky. Arithmetical pattern problems relating to divisibility by r th powers. *Proceedings of the London Mathematical Society*, s2-50(1):497–508, 1948.
- [3] RH research program. Bounded constraint-graph transducer certificates, 2026. Repository paper RH-372, frozen release.
- [4] RH research program. Composite-clock phase selectors and an improved mobius capacity floor, 2026. Repository paper RH-373, frozen release.
- [5] RH research program. Eight-run reduction and the cyclic pair-ledger obstruction, 2026. Repository paper RH-371, frozen release.
- [6] RH research program. Möbius orthogonality, adaptive encoding, and parry covariance, 2026. Repository paper RH-366, frozen release.
- [7] RH research program. Square clocks, run recurrences, and an euler-product floor for mobius distance-two capacity, 2026. Repository paper RH-374, frozen release.